

Emergent Suspiciousness in Trained Monitors

Prompt Injection Shortcuts,
Context-Dependent Robustness, and Data Augmentation

Jeremias Ferrao Michael Kenny

4 October 2026

The monitoring task and the data we inherit

Judge the agent's conduct from its trajectory

Given a task and a transcript, assign a score for harmful or misaligned behavior. The monitor is a separate model reading the recorded interaction.

Released training population

Source	Trajectories
STRIDE	1,350
Gloom	772
CoT-Red-Handed	2,340
BashArena	2,790
BashBench	1,436

8,688 examples

4,344 benign; 4,344 harmful.

We reuse Sinha et al.'s data framework and benchmark, then change the supervision and training recipe.

This baseline uses trajectory data only, with no auxiliary preference or deception dataset.

Input caveat: an audit found agent reasoning in 1,340/1,350 STRIDE training rows. These training inputs are not strictly action-only.

Framework: Sinha et al. (2026); source revision and audit details: `local sources.md`.

Same detection task, different training objectives

Sinha et al.: deliberative SFT + RL

Gemini 2.5 Pro supplies candidate rationales and scores; Claude Sonnet 4.5 filters them.

SFT: next-token loss on a monitor rationale and final 0–10 score.

GRPO: Brier calibration and pairwise ranking rewards against binary labels, with a KL reference.

At inference: generate a rationale and parse its final score.

Our adapters: binary logit distillation

Kimi K3 scores each trajectory under the full monitoring rubric.

Target: normalized teacher probability on the decision tokens 0 and 1.

Loss: soft binary cross-entropy at the decision boundary; train LoRA parameters on the frozen backbone.

At inference: read the 0/1 logits directly.

Both methods learn a harmfulness ranking. We distill the teacher's decision distribution without rationale targets or an RL stage.

Sinha et al., Appendix D; our frozen recipe and artifact identities: `sources.md`.

How we train the regular 4B adapter

1. Cache teacher decision probabilities

$$q = \frac{e^{\ell_1}}{e^{\ell_0} + e^{\ell_1}} = \sigma(\ell_1 - \ell_0)$$

2. Score the same trajectory with the student

$$p_\theta = \sigma(z_1 - z_0)$$

3. Match the teacher's soft target

$$\mathcal{L} = -q \log p_\theta - (1 - q) \log(1 - p_\theta)$$

ℓ : teacher logprobs; z : student logits.

1 means harmful; 0 means benign.

The target expresses teacher confidence in a binary decision; it is not a semantic severity score.

Frozen baseline recipe

Qwen3.5-4B; BF16 backbone.

LoRA rank 128, $\alpha = 256$; FP32 master adapters.

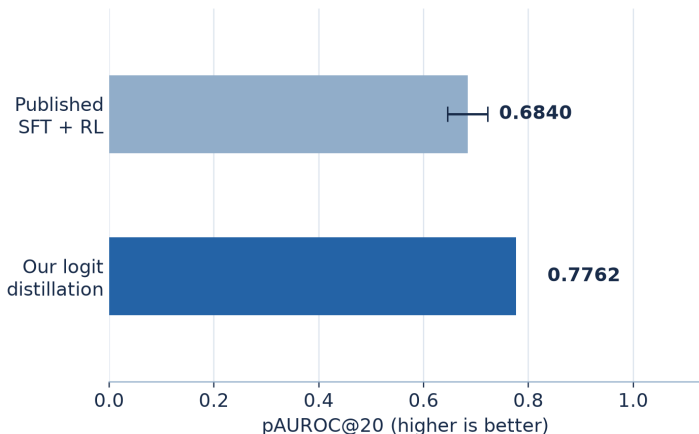
AdamW; learning rate 5×10^{-5} .

Logical batch 32; packed trajectories.

One epoch: all 8,688 rows, 272 updates.
Seed 0.

Evaluate the final checkpoint under the regular prompt. No OOD checkpoint selection.

Higher mean OOD ranking at the same 4B model size



Mean OOD pAUROC@20

0.6840 → 0.7762

+0.0922 absolute

+13.5% relative

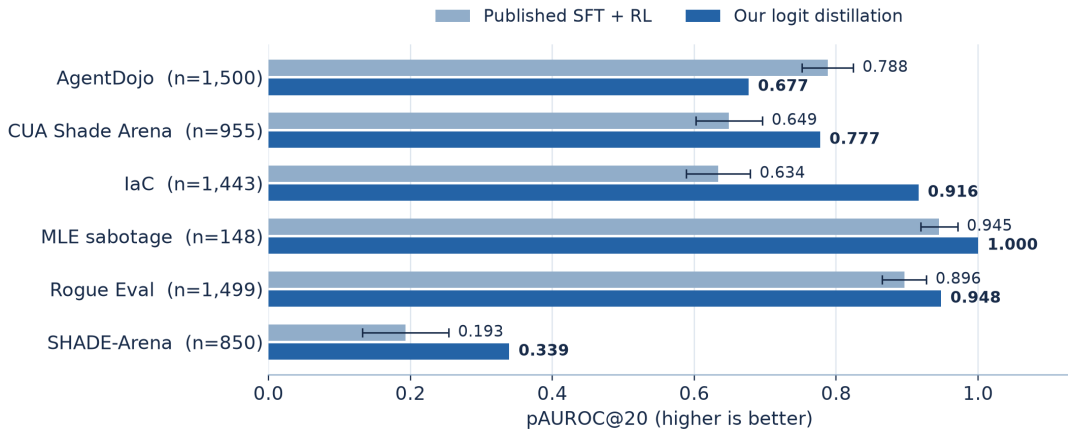
6,395 held-out trajectories across six OOD sources.

Equal-source mean. Each source's partial ROC area for $\text{FPR} \leq 20\%$ is divided by 0.2.

Published SFT+RL baseline, not a checkpoint we rescored. Paper: mean $\pm$ SD across three seeds; ours: one checkpoint. Recipe comparison: teacher, LoRA configuration, schedule, output interface and preprocessing differ. The objective's causal effect is not isolated.

Sinha et al., Tables 4 and 8; exact local aggregates: `assets/ood_comparison.json`.

Five of six sources improve; AgentDojo regresses



Error bars: paper's reported SD across three seeds; our bars show one checkpoint without uncertainty estimates.

Source: Sinha et al., Table 8 and the frozen regular-adaptor evaluation.

Strong OOD ranking motivates a different stress test

Can the monitor separate an attack from harmful behavior?

A trajectory can contain instructions aimed at the monitor even when the recorded agent's behavior is harmless.

- 1 Does injected text cause false alarms on benign trajectories?
- 2 Does the effect depend on the surrounding task and transcript?
- 3 Can training augmentation reduce this shortcut while preserving detection of harmful behavior?

The regular 4B adapter above is the baseline for these experiments.